

HR Analytics Dashboard Report

1. Problem Statement

Employee attrition is one of the major challenges faced by organizations, resulting in increased recruitment costs, reduced productivity, knowledge loss, and lower employee morale. HR departments often struggle to identify the factors contributing to employee turnover due to the absence of centralized analytics and real-time workforce insights.

Organizations require a comprehensive analytical solution to understand workforce demographics, employee satisfaction, compensation, work-life balance, overtime patterns, and departmental performance. Without data-driven insights, it becomes difficult to identify high-risk employees, improve retention strategies, and make informed workforce planning decisions.

The primary objective of this project is to analyze HR data using Tableau to identify the key factors influencing employee attrition and provide actionable insights for improving employee retention and organizational performance.

2. Dataset Overview

The HR Analytics dataset contains employee information collected from various organizational departments.

Key Attributes

Employee Information

- Employee ID
- Employee Number
- Age
- Age Group
- Gender
- Marital Status

Job Details

- Department
- Job Role
- Job Level
- Business Travel

- Total Working Years
- Years at Company
- Years in Current Role
- Years Since Last Promotion
- Years With Current Manager

Compensation

- Monthly Income
- Salary Slab
- Daily Rate
- Hourly Rate
- Stock Option Level
- Percent Salary Hike

Employee Satisfaction

- Job Satisfaction
- Environment Satisfaction
- Relationship Satisfaction
- Work-Life Balance
- Job Involvement
- Performance Rating

Attrition Information

- Attrition
- Overtime

3. Data Preprocessing Steps

The following preprocessing steps were performed before building the Tableau dashboards:

- Removed duplicate records.
- Verified that no critical missing values were present.
- Checked and corrected data types for numerical and categorical columns.

- Removed unnecessary columns with constant values:
 - EmployeeCount
 - Over18
 - StandardHours
 - Verified employee identifiers.
 - Standardized column names where required.
 - Created calculated fields for:
 - Attrition Rate
 - Attrition Count
 - Active Employees
 - Average Monthly Income
 - Average Experience
 - Attrition Category
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4. Dashboards Created

4.1 Workforce Overview Dashboard

Purpose

Provide a high-level overview of the organization's workforce.

KPI Cards

- Total Employees
- Active Employees
- Attrition Count
- Average Monthly Income

Visualizations

- Employee Distribution by Department
- Employee Distribution by Job Role
- Employees by Age Group

- Gender Distribution
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4.2 Attrition Analysis Dashboard

Purpose

Identify departments, job roles, and employee groups with the highest employee turnover.

KPI Cards

- Attrition Rate
- Average Years at Company
- Average Job Satisfaction
- Average Work-Life Balance

Visualizations

- Attrition by Department
 - Attrition by Job Role
 - Attrition by Salary Slab
 - Overtime vs Attrition
 - Business Travel vs Attrition
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4.3 Employee Satisfaction & Performance Dashboard

Purpose

Analyze employee engagement and workplace factors affecting retention.

KPI Cards

- Average Job Satisfaction
- Average Environment Satisfaction
- Average Relationship Satisfaction
- Average Performance Rating

Visualizations

- Job Satisfaction by Department
- Work-Life Balance vs Attrition

- Monthly Income by Job Role
 - Total Working Years vs Monthly Income
 - Environment Satisfaction Distribution
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4.4 Executive Summary Dashboard

Purpose

Provide senior management with a consolidated overview of workforce performance and attrition trends.

KPI Cards

- Total Employees
- Attrition Rate
- Average Monthly Income
- Average Experience

Visualizations

- Attrition Trend by Years at Company
 - Top Job Roles with Highest Attrition
 - Department-wise Average Salary
 - Employee Distribution by Marital Status
 - Stock Option Level vs Attrition
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5. Tableau Calculated Fields Used

5.1 Basic Calculations

Total Employees

COUNT([EmpID])

Attrition Count

SUM(

IF [Attrition]="Yes"

THEN 1

ELSE 0

END

)

Active Employees

COUNT([EmpID]) -

SUM(

IF [Attrition]="Yes"

THEN 1

ELSE 0

END

)

5.2 Derived Calculations

Attrition Rate

SUM(

IF [Attrition]="Yes"

THEN 1

ELSE 0

END

)

/COUNT([EmpID])

Average Monthly Income

AVG([MonthlyIncome])

Average Experience

AVG([TotalWorkingYears])

5.3 Custom Categorization

Attrition Category

IF [YearsAtCompany] <=2 THEN "Early Exit"

ELSEIF [YearsAtCompany]<=5 THEN "Mid Career"

ELSE "Long Term"

END

6. Key Insights & Findings

- The Research & Development department has the largest workforce, while attrition is concentrated in specific departments and job roles.
 - Employees working overtime show a noticeably higher attrition rate compared to those who do not work overtime.
 - Lower salary slabs experience greater employee turnover than higher salary groups.
 - Employees with lower job satisfaction and work-life balance scores are more likely to leave the organization.
 - Employees with fewer years at the company exhibit higher attrition, indicating that retention challenges are more significant during the early stages of employment.
 - Monthly income generally increases with total working experience and job level.
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7. Assumptions & Business Recommendations

Assumptions

- The dataset accurately represents employee records.
 - Employee satisfaction ratings reflect actual employee feedback.
 - No external economic or organizational events influencing attrition were considered.
 - Employee records are complete and up to date.
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Recommendations

- Introduce targeted retention programs for employees during their first few years of service.
- Monitor overtime policies to reduce employee burnout.
- Improve work-life balance through flexible work arrangements and wellness initiatives.
- Review compensation strategies for lower salary groups to improve retention.
- Conduct regular employee satisfaction surveys and address concerns proactively.
- Focus retention efforts on departments and job roles with consistently high attrition.

8. Final Conclusion (HR Analytics Report Summary)

The HR Analytics Dashboard provides a comprehensive overview of workforce demographics, employee satisfaction, compensation, and attrition patterns. Through interactive Tableau dashboards and calculated fields, HR managers can identify the primary drivers of employee turnover and monitor workforce performance effectively.

The analysis highlights the importance of work-life balance, job satisfaction, compensation, overtime, and employee experience in influencing attrition. By implementing targeted retention strategies based on these insights, organizations can reduce turnover, improve employee engagement, optimize workforce planning, and support long-term organizational growth.